CONCEPT NOTE AND TECHNICAL PROPOSAL · PROBLEM STATEMENT 3: LOST-PERSON IDENTIFICATION AND REUNIFICATION ASSISTANCE

Band, trail and the right desk: reuniting lost children and elders in minutes

Children and senior citizens get a simple band at entry. The roadside nodes already installed for crowd alerts quietly note anonymous band sightings. When a family reports someone missing, by desk, missed call or volunteer, the system finds the last-seen place and trail and sends both sides to the same help desk. No faces, no names, no smartphone needed.

Team IIIT Hyderabad **Challenge** RTIH × RMC Innovation-Led Solutions for Godavari Pushkaralu 2027

Date November 2026 **Demo video** [link to PS03 video]

SUMMARY

At mass gatherings, separations are common and slow to resolve: the family waits at one place, the found child at another, and announcements over loudspeakers rarely connect them. Our design fixes the meeting problem first. At entry, the family is told which help desk to go to if they get separated, and a found child or elder is taken to **that same desk**, the one nearest where the separation happened. A low-cost Bluetooth band adds a **last-seen place and trail**, read from the nodes only after a report, so volunteers know where to look. Reports work from any phone through a **missed call**. In simulation this halves the typical time to reunite.

37 → 17 min
median time to reunite, today's practice vs our design (3,000 simulated cases each)

28% → 79%
of families reunited within 30 minutes

≤ 29 min
median even if only 30% of families follow the desk advice

0
faces, names or phone numbers on the band; sightings deleted after 60 min

The reunion results come from a simulation with stated behaviour assumptions (§4), not field data. Each section below names the evaluation criterion it answers.

1 · THE PROBLEM · PROBLEM–SOLUTION ALIGNMENT

Many get separated, few carry smartphones, and faces are off limits

- **Scale.** At Maha Kumbh 2025, about **54,000** separated people were reunited with their families, most of them women and elderly pilgrims [1]. Godavari Pushkaralu 2015 drew about **11 crore** bathers over 12 days [2]; for 2027 the state expects up to 10 crore [7].
- **Phones.** Many rural pilgrims carry no smartphone, and often only one family member has any phone. A child may be with a relative or a neighbour, not a parent.
- **No facial recognition.** The challenge rules exclude facial recognition without explicit approval (§10). Kumbh 2025's digital lost-and-found centres relied on it [1]; Pushkaralu needs a method that does not.
- **Today's practice.** Loudspeaker announcements and paper registers. The family and the found person often wait at



Mauni Amavasya crowd, Kumbh Mela 2019. Photo: Michael T Balonek, CC BY-SA 4.0, Wikimedia Commons.

different places. In our simulation the median time to reunite is **37 minutes**, and only 28% are reunited within 30 minutes.

2 · OUR SOLUTION · INNOVATION AND ORIGINALITY

Five steps, built on the crowd-alert nodes already on site

Step	What happens
1. Band at entry	**Only children and senior citizens** get a band, with a QR code, the helpline and a case code; no phone number is printed on it. The family gives **any reachable number** (a feature phone, a relative, the group leader) and is told **which help desk to go to** if separated.
2. Quiet sightings	The gate and road nodes from our PS-01 crowd system note when a band passes. They store only **anonymous IDs that change every few minutes**, keep them for 60 minutes, and send nothing to a central server.
3. Report	The family reports at a help desk, by a **missed call** from any phone (the system calls back), or through a volunteer. Only then does the desk ask the nodes about that band, and get back its **last-seen place and trail**. The band switches to "lost" mode and beacons faster.
4. Meet at the right desk	The found person is taken to the **desk nearest where they were separated**, the same desk the family was told to go to, so both sides converge on one place.
5. Exit watch	If a reported band passes an exit gate, the post there is alerted. This covers a child being led away by a stranger.

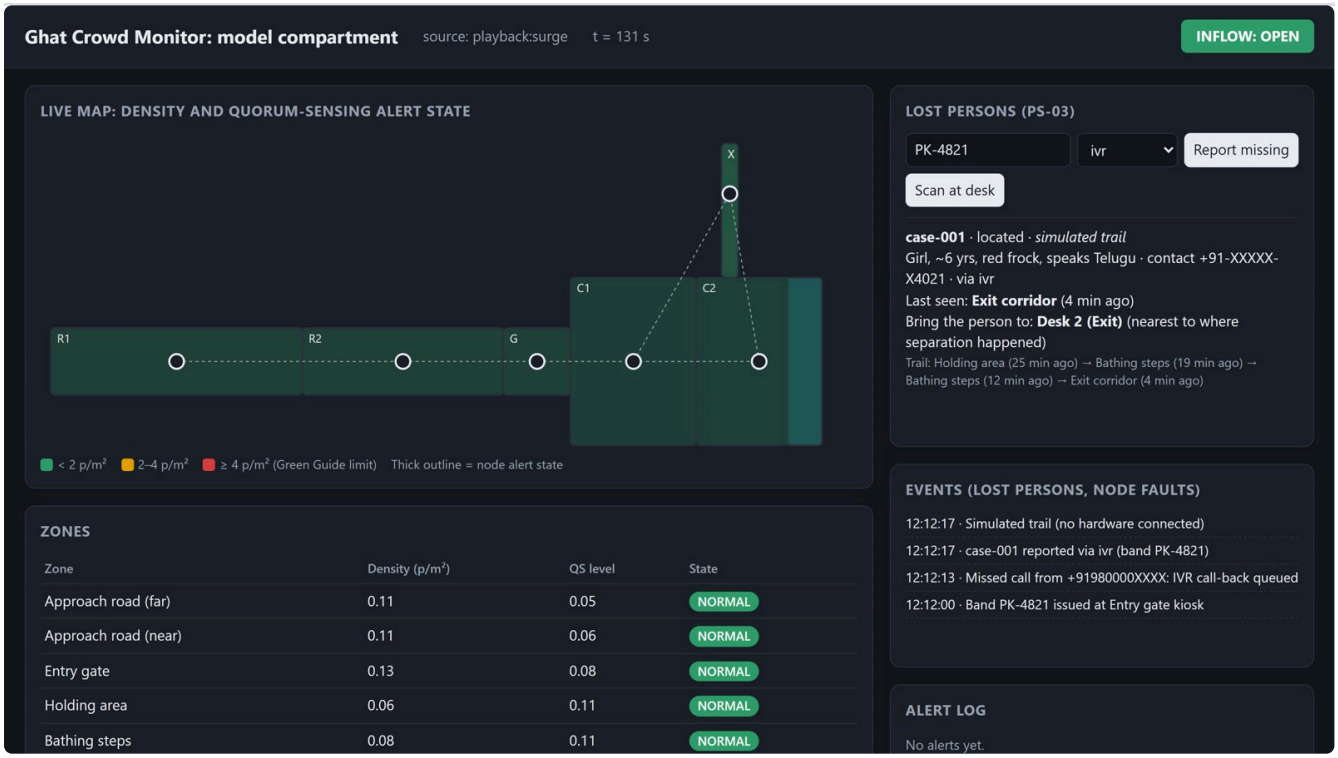


Figure 1. Our prototype dashboard after a missed-call report for band PK-4821: last seen in the exit corridor 4 minutes ago, the trail through the holding area and bathing steps, and "Bring the person to: Desk 2". In this demo the trail is simulated; no boards are connected.

What is new: reunion is treated as a **meeting** problem (converge-at-desk) as well as a search problem, and the search uses privacy-preserving band sightings from infrastructure that is already installed for crowd safety, with no cameras and no app.

3 · TECHNICAL ARCHITECTURE · TECHNICAL FEASIBILITY, SCALABILITY AND CYBERSECURITY

Opaque IDs on the roadside, keys only in the registry

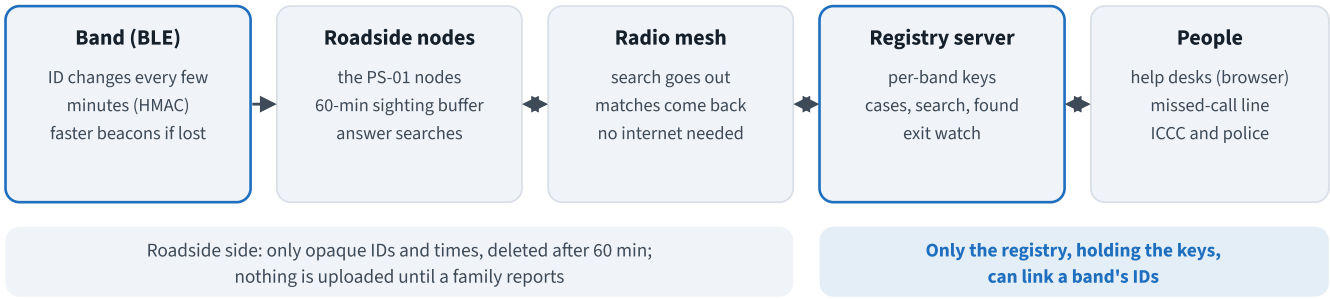


Figure 2. System architecture. A search is the only time sightings leave a node, and only for the reported band.

Layer	Technology	Status
Band	ESP32 tag in the prototype; a low-cost nRF52 BLE wristband in deployment. ID = HMAC-SHA256(per-band key, time slot); the radio address rotates with it	Firmware done; not yet field-tested
Nodes	The PS-01 ESP32 nodes: Bluetooth scan every second; 60-minute buffer of opaque IDs with time and signal strength	Done in firmware
Search	The registry computes the band's IDs for the last hour and broadcasts them over the mesh (LoRa in deployment); matching nodes reply with zone and time	Done over ESP-NOW; the demo trail is simulated
Server	Python FastAPI: issue band, report (desk, missed call, volunteer), cases, found scan with notification, exit watch; REST with OpenAPI documentation	Done; end-to-end tests pass
Missed-call line	A missed-call number from a telecom or IVR provider that calls our documented endpoint	Endpoint done; line to be procured

- **Built in:** a separate secret key for every band (a test caught two bands sharing a key, now fixed); nodes cannot link IDs; sightings expire after 60 minutes.
- **Planned for the pilot:** TLS on all traffic; role-based logins for help desks; an audit log of every search; the registry encrypted at rest; consent records kept with each band; retention set by RMC policy; hosting in the RMC or state data centre following CERT-In guidelines.

4 · EVIDENCE · TECHNICAL FEASIBILITY AND EXPECTED IMPACT

Meeting at the right desk is what halves the wait

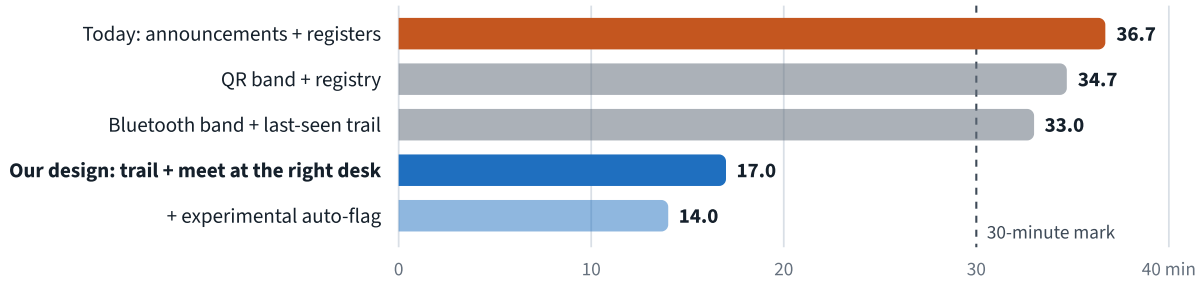


Figure 3. Median time to reunite, 3,000 simulated cases per approach. The band and trail alone save little; sending both sides to the same desk is what halves the wait.

Approach	Median	90th percentile	Within 30 min	Within 60 min
Today: announcements + paper registers	36.7 min	74.0	28%	84%
QR band + registry	34.7	52.7	33%	92%
Bluetooth band + last-seen trail	33.0	43.7	36%	99%
Our design: trail + meet at the right desk	**17.0**	**37.7**	**79%**	**99%**
+ experimental auto-flag (family phone learnable)	14.0	35.0	82%	99.6%

- **Robust to family behaviour.** Across family compliance from 30% to 90% and other assumption changes, our design's median stays at 14–29 minutes, against 37–38 for today's practice. It wins in every combination tested.
- **Assumptions, stated openly:** parents notice after 2–10 min and search nearby for 10–30 min; a lost child drifts to a neighbouring zone at 0.3 per minute; bystanders find a child at 1 per 20 min and alerted volunteers at 1 per 4 min; 70% of families follow the desk advice; nodes detect a passing band 90% of the time. Site model: 60 zones of 40 m with 6 help desks.
- **Auto-flag is an extension, not the core.** For families whose phone the band can learn, the band can flag a separation on its own (median 14 min), but in dense crowds it still raises too many false call-outs, so we will tune it during the pilot.

5 · PRIVACY AND SECURITY · DATA PROTECTION AND COMPLIANCE

- **No faces, no names, no printed phone number.** The band carries a QR code, the helpline and a case code.
- **Strangers can't follow a band.** Its Bluetooth ID is HMAC-SHA256 of a per-band secret key and the time slot, and its radio address changes at the same moment. Only the registry, holding the key, can link the IDs, and only after a report.
- **Data stays local and expires.** Nodes keep sightings for 60 minutes and upload nothing centrally. A search runs only when a family reports.
- **Consent.** Guardian consent is taken when the band is issued, in line with the Digital Personal Data Protection Act, 2023, which requires verifiable parental consent for children's data [3]. We will review the wording with RMC's legal team before the pilot.
- **Planned for the pilot:** role-based access for help desks, an audit log of every search, and case records kept only as long as RMC policy requires.

6 · PROTOTYPE · IMPLEMENTATION READINESS

Part	Status
Band firmware (ESP32 tag role)	Rotating HMAC IDs and rotating radio address; faster beacons when lost; compiles (12% of RAM). **Not yet field-tested.**
Node firmware	The PS-01 node keeps a 60-minute sighting buffer, answers searches and relays the "lost" command.
Backend API	Issue band, report (desk / missed call / volunteer), search, found scan with notification, exit watch, case list. End-to-end tests pass.
Dashboard	Case panel with trail, designated desk, exit warning and event log (Figure 1).
Demo (12 Nov)	We walk a band past the nodes, report it by missed call, and the dashboard shows the trail and the desk to go to.

7 · FIT WITH PUSHKARALU REQUIREMENTS · SCALABILITY AND INTEROPERABILITY

Requirement	How the design meets it
Pilgrims without smartphones	Missed-call reporting from any phone; desks and volunteers can file reports; the meet-at-desk step works with no device at all.
Multilingual access	Colour-coded desks and the printed band need little reading; desk scripts and call-back prompts to be prepared in Telugu, Hindi and English for the pilot.
Field usability	Help desks use any browser; volunteers need only the case code and the desk number.
Reliability	Sightings and searches run on the node mesh, so a search works even when the mobile network is congested.
Interoperability	Documented REST API; cases can be shared with the ICCC and police lost-and-found registers.
Scale and cost	Bands only for children and senior citizens; nodes are shared with PS-01, so no extra infrastructure.
Post-event use	Temples, fairs, school trips and other mass gatherings in the district.

8 · PLAN AND INDICATIVE COST · IMPLEMENTATION AND COST-EFFECTIVENESS

When	What	Measured
Nov 2026 – Jan 2027	Agree help-desk locations, band issuance points and the missed-call number; legal review of consent	Issuance time per family
Feb – Mar 2027	**Live pilot at Maha Shivaratri**, alongside PS-01	Time to reunite, % within 30 min, node detection rate, false exit alerts
Mar – Jun 2027	Scale to all entry points; volunteer and police drills	Drill reunion times
Jun – Jul 2027	12-day operation with the ICCC and police	Live KPIs

- **Bands:** retail Bluetooth wristband beacons cost about **₹1,300–1,530** each in India [4, 5]; overseas bulk listings start near US$8 [6]. Reusing bands (collected at exits) or giving printed QR-only bands to adults with phones would cut the cost; the right mix is a decision for RMC.
- **Nodes:** no extra cost; the PS-01 nodes do the sighting.
- **Desks and calls:** any PC or tablet with a browser; a missed-call number from a telecom provider (to be quoted).

Indicative only: public retail listings (November 2026) before GST and bulk discounts; to be replaced by vendor quotes.

9 · LIMITATIONS AND RISKS

- **The reunion results depend on assumed behaviour.** No field data was used; the pilot will measure it.
- The radio model comes from published measurements, not our own. Whether ordinary Android phones advertise over Bluetooth (needed only for the auto-flag extension) will be measured at the pilot.
- A parent with no device cannot be detected by any radio system; for them, meeting at the right desk is what makes the difference, and it needs no device.
- Band issuance at busy entry points needs staff and a fast desk process; the cost of bands is the main budget item.

10 · WHAT WE NEED FROM RMC AND RTIH

1. Planned help-desk and band-issuance locations, and the Kotilingala model-compartment layout.
2. A missed-call number or IVR line, and the link to police lost-and-found procedures.

3. Pilot permission at Maha Shivaratri, with volunteers and police.

11 · TEAM

Team IIIT Hyderabad. [Member names, roles, programme and year] · Contact: [name, email, phone]

Also submitted: PS-01 route-aware crowd alerts, whose nodes this design reuses.

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